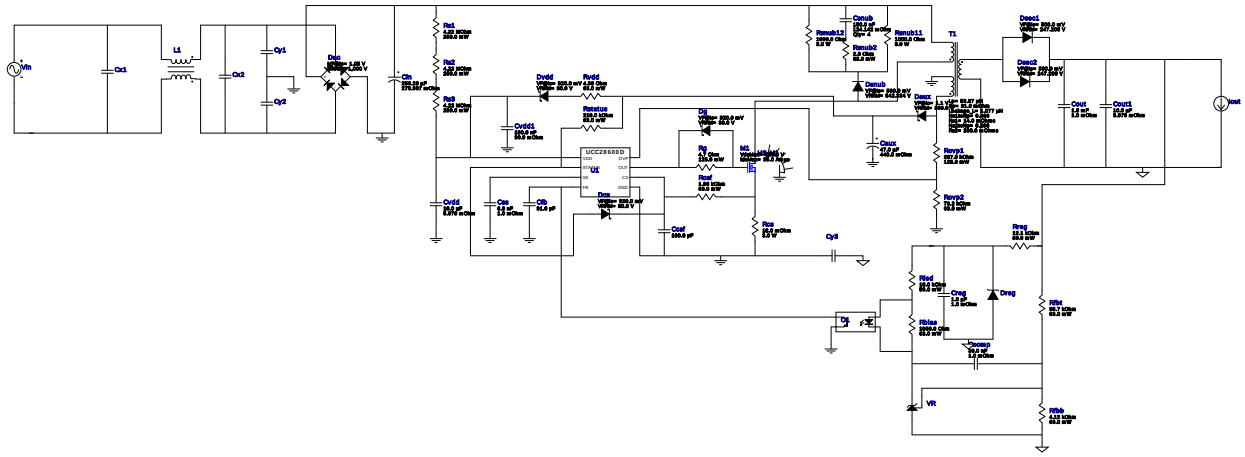
WEBENCH® Design Report

VinMin = 220.0V
VinMax = 265.0V
Vout = 24.0V
Iout = 30.0A

Device = UCC28600DR
Topology = Flyback
Created = 2025-09-27 08:05:04.648
BOM Cost = NA
BOM Count = 51
Total Pd = 140.68W

Design : 1 UCC28600DR
UCC28600DR 220V-265V to 24.00V @ 30A

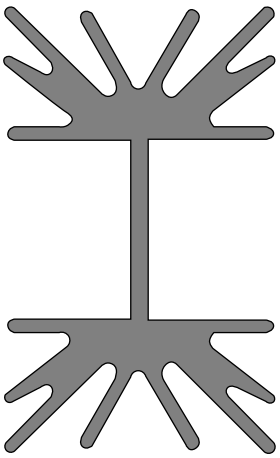


1. The EMI filter shown in the schematic is a placeholder. It has not yet been designed for the application.

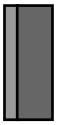
Electrical BOM

Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
Caux	Nichicon	UUD1V470MCL1GS Series= uD	Cap= 47.0 uF ESR= 440.0 mOhm VDC= 35.0 V IRMS= 230.0 mA	1	$0.13	SM_RADIAL_6.3AMM 80 mm²
Ccomp	MuRata	GRM155R71A393KA01D Series= X7R	Cap= 39.0 nF ESR= 1.0 mOhm VDC= 10.0 V IRMS= 0.0 A	1	$0.01	0402 3 mm²
Ccsf	Kemet	C0201C101K3GACTU Series= C0G/NP0	Cap= 100.0 pF VDC= 10.0 V IRMS= 0.0 A	1	$0.01	0201 2 mm²
Cfb	MuRata	GRM0335C1E910JA01D Series= C0G/NP0	Cap= 91.0 pF VDC= 25.0 V IRMS= 0.0 A	1	$0.01	0201 2 mm²
Cin	CUSTOM	CUSTOM Series= ?	Cap= 256.28 uF ESR= 278.91 mOhm VDC= 401.0 V IRMS= 9.6402 A	1	NA	CUSTOM 0 mm²
Cout	CUSTOM	CUSTOM Series= ?	Cap= 1.8 mF ESR= 1.0 mOhm VDC= 60.0 V	1	NA	CUSTOM 0 mm²
Cout1	Taiyo Yuden	MSASU32MSB5106KPNA01 Series= X5R	Cap= 10.0 uF ESR= 3.979 mOhm VDC= 50.0 V IRMS= 3.4821 A	1	$0.16	1210 15 mm²
Creg	Taiyo Yuden	EMK107B7105KA-T Series= X7R	Cap= 1.0 uF ESR= 1.0 mOhm VDC= 16.0 V IRMS= 0.0 A	1	$0.01	0603 5 mm²
Creg	Taiyo Yuden	EMK107B7105KA-T Series= X7R	Cap= 1.0 uF ESR= 1.0 mOhm VDC= 16.0 V IRMS= 0.0 A	1	$0.01	0603 5 mm²

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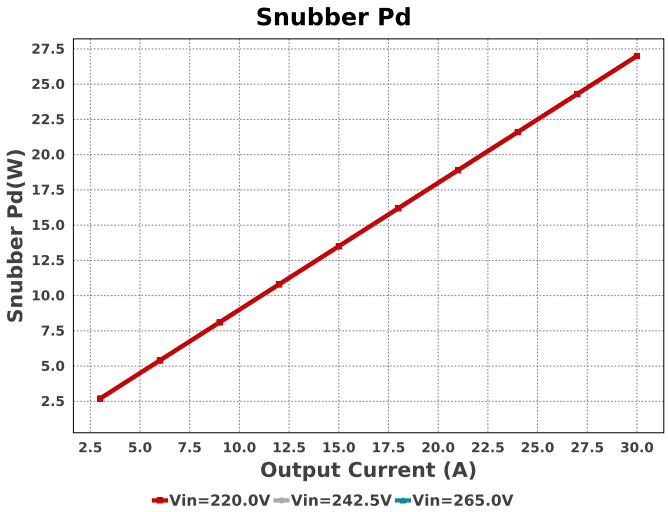
Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
Csnub	Taiyo Yuden	MCASQ32NSB7154KTNA01 Series= X7R	Cap= 150.0 nF ESR= 124.142 mOhm VDC= 250.0 V IRMS= 699.24 mA	4	$0.14	 1210 15 mm²
Css	MuRata	GRM033R71A682KA01D Series= X7R	Cap= 6.8 nF ESR= 1.0 mOhm VDC= 10.0 V IRMS= 0.0 A	1	$0.01	 0201 2 mm²
Cvdd	Taiyo Yuden	MSASG21GBB5106MTNA01 Series= X5R	Cap= 10.0 uF ESR= 5.979 mOhm VDC= 35.0 V IRMS= 1.88442 A	1	$0.17	 0805 7 mm²
Cvdd1	MuRata	GRM188R71E104KA01D Series= X7R	Cap= 100.0 nF ESR= 30.0 mOhm VDC= 25.0 V IRMS= 1.51 A	1	$0.01	 0603 5 mm²
Dac	Diodes Inc.	GBJ2510-F	VF@Io= 1.05 V VRRM= 1,000.0 V	1	$1.18	GBJ 211 mm²
Daux	SMC Diode Solutions	ST1300ATR	VF@Io= 1.1 V VRRM= 300.0 V	1	$0.12	 SMA 37 mm²
Dcs	Toshiba	CMS06	VF@Io= 320.0 mV VRRM= 30.0 V	1	$0.20	 M-FLAT 19 mm²
Dg	Toshiba	CMS06	VF@Io= 320.0 mV VRRM= 30.0 V	1	$0.20	 M-FLAT 19 mm²
Dreg	Diodes Inc.	SMBJ5.0A-13-F	Zener	1	$0.10	 SMB 44 mm²
Dreg	Diodes Inc.	SMBJ5.0A-13-F	Zener	1	$0.10	 SMB 44 mm²
Dsec1	CUSTOM	CUSTOM	VF@Io= 500.0 mV VRRM= 247.208 V	1	NA	CUSTOM 0 mm²
Dsec2	CUSTOM	CUSTOM	VF@Io= 500.0 mV VRRM= 247.208 V	1	NA	CUSTOM 0 mm²
Dsnub	CUSTOM	CUSTOM	VF@Io= 500.0 mV VRRM= 542.334 V	1	NA	CUSTOM 0 mm²
Dvdd	Toshiba	CMS06	VF@Io= 320.0 mV VRRM= 30.0 V	1	$0.20	 M-FLAT 19 mm²
HS_M1	Aavid	529802B02500G	Heatsink	1	$6.16	 529802 1203 mm²

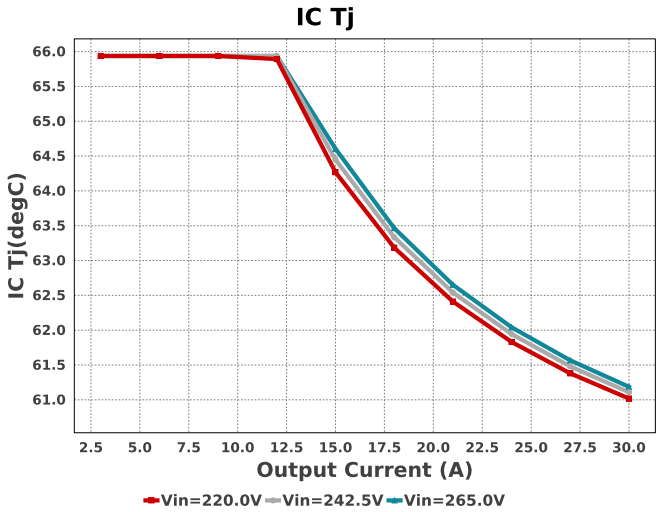
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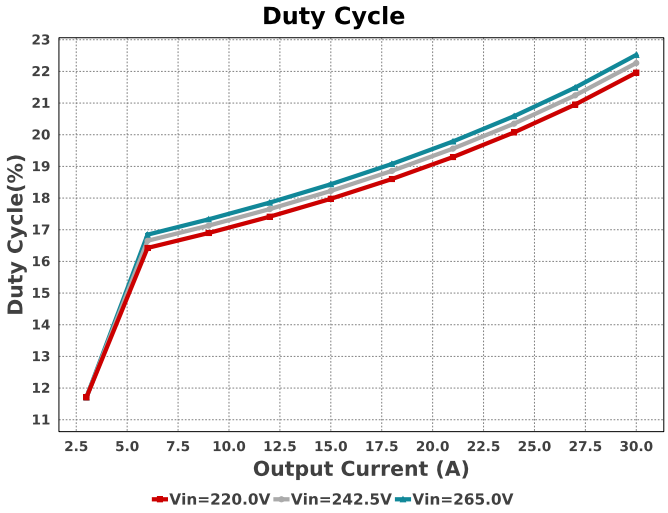
Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
M1	STMicroelectronics	STP45N65M5	VdsMax= 650.0 V IdsMax= 35.0 Amps	1	$6.03	 TO-220AB 79 mm²
O1	Vishay-Semiconductor	TCMT1107	Optocoupler	1	$0.19	 SOP-4 44 mm²
Rbias	Vishay-Dale	CRCW04021K00FKED Series= CRCW..e3	Res= 1000.0 Ohm Power= 63.0 mW Tolerance= 1.0%	1	$0.01	 0402 3 mm²
Rcs	Susumu Co Ltd	KRL7638-C-R018-F-T1 Series= KRL	Res= 18.0 mOhm Power= 3.0 W Tolerance= 1.0%	1	$0.56	 3015 56 mm²
Rcsf	Vishay-Dale	CRCW04021K96FKED Series= CRCW..e3	Res= 1.96 kOhm Power= 63.0 mW Tolerance= 1.0%	1	$0.01	 0402 3 mm²
Rfbb	Vishay-Dale	CRCW04024K12FKED Series= CRCW..e3	Res= 4.12 kOhm Power= 63.0 mW Tolerance= 1.0%	1	$0.01	 0402 3 mm²
Rfbt	Vishay-Dale	CRCW040235K7FKED Series= CRCW..e3	Res= 35.7 kOhm Power= 63.0 mW Tolerance= 1.0%	1	$0.01	 0402 3 mm²
Rg	Yageo	RC0805FR-074R7L Series= ?	Res= 4.7 Ohm Power= 125.0 mW Tolerance= 1.0%	1	$0.01	 0805 7 mm²
Rled	Yageo	RC0201FR-0710KL Series= ?	Res= 10.0 kOhm Power= 50.0 mW Tolerance= 1.0%	1	$0.01	 0201 2 mm²
Rovp1	Panasonic	ERJ-6ENF2673V Series= ERJ-6E	Res= 267.0 kOhm Power= 125.0 mW Tolerance= 1.0%	1	$0.01	 0805 7 mm²
Rovp2	Vishay-Dale	CRCW040273K2FKED Series= CRCW..e3	Res= 73.2 kOhm Power= 63.0 mW Tolerance= 1.0%	1	$0.01	 0402 3 mm²
Rreg	Yageo	RC0201FR-0712K1L Series= ?	Res= 12.1 kOhm Power= 50.0 mW Tolerance= 1.0%	1	$0.01	 0201 2 mm²
Rreg	Yageo	RC0201FR-0712K1L Series= ?	Res= 12.1 kOhm Power= 50.0 mW Tolerance= 1.0%	1	$0.01	 0201 2 mm²
Rs1	Vishay-Dale	CRCW12064M22FKEA Series= CRCW..e3	Res= 4.22 MOhm Power= 250.0 mW Tolerance= 1.0%	1	$0.01	 1206 11 mm²
Rs2	Vishay-Dale	CRCW12064M22FKEA Series= CRCW..e3	Res= 4.22 MOhm Power= 250.0 mW Tolerance= 1.0%	1	$0.01	 1206 11 mm²
Rs3	Vishay-Dale	CRCW12064M22FKEA Series= CRCW..e3	Res= 4.22 MOhm Power= 250.0 mW Tolerance= 1.0%	1	$0.01	 1206 11 mm²
Rsnub11	Vishay-Bccomponents	PR03000201001JAC00 Series= ?	Res= 1000.0 Ohm Power= 3.0 W Tolerance= 5.0%	1	$0.11	 PR03 197 mm²
Rsnub12	Vishay-Bccomponents	PR03000201001JAC00 Series= ?	Res= 1000.0 Ohm Power= 3.0 W Tolerance= 5.0%	1	$0.11	 PR03 197 mm²
Rsnub2	Vishay-Dale	CRCW04022R80FKED Series= CRCW..e3	Res= 2.8 Ohm Power= 63.0 mW Tolerance= 1.0%	1	$0.01	 0402 3 mm²

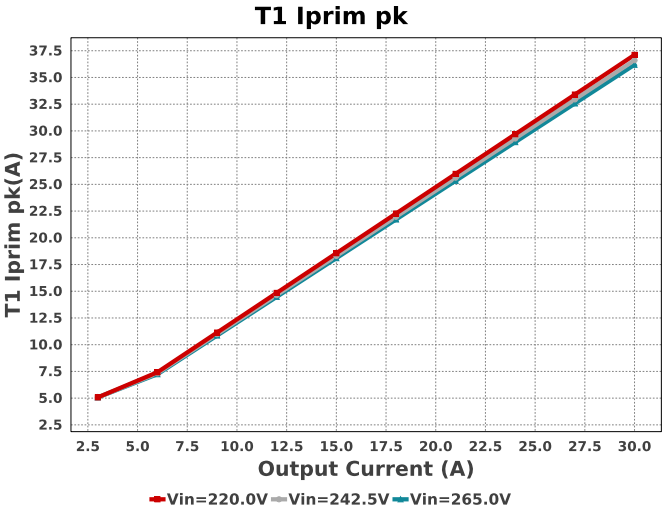
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Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
Rstatus	Vishay-Dale	CRCW0402210KFKED Series= CRCW..e3	Res= 210.0 kOhm Power= 63.0 mW Tolerance= 1.0%	1	$0.01	0402 3 mm^2
Rvdd	Vishay-Dale	CRCW04024R99FKED Series= CRCW..e3	Res= 4.99 Ohm Power= 63.0 mW Tolerance= 1.0%	1	$0.01	0402 3 mm^2
T1	CUSTOM	CUSTOM	Lp= 53.87 µH Rp= 31.0 mOhm Leakage_L= 1.077 µH Ns1toNp= 466.431 m Rs1= 14.0 mOhms Ns2toNp= 0.298 Rs2= 200.0 mOhms	1	NA	CUSTOM 0 mm^2
U1	Texas Instruments	UCC28600DR	Switcher	1	$0.36	 D0008A 57 mm^2
VR	Texas Instruments	TL431IDBVR	Voltage References	1	$0.06	 R-PDSO-G3 16 mm^2



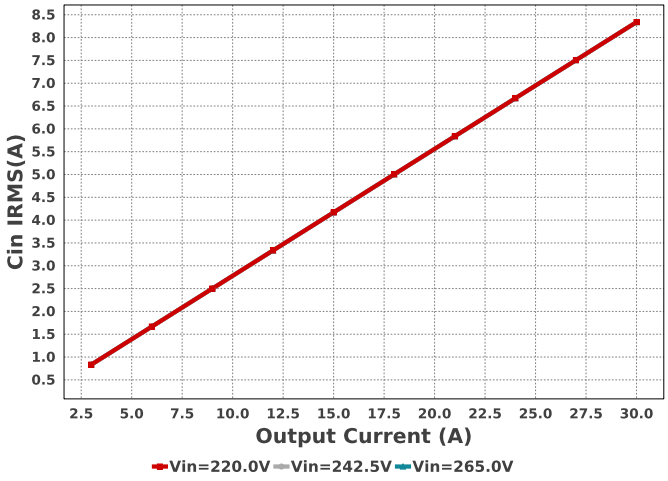




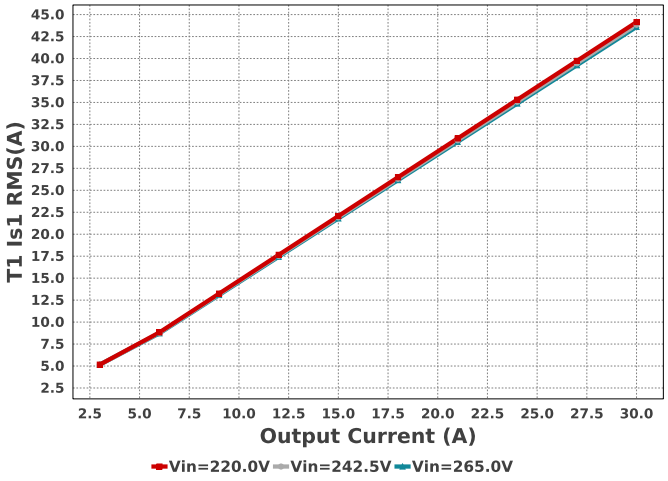


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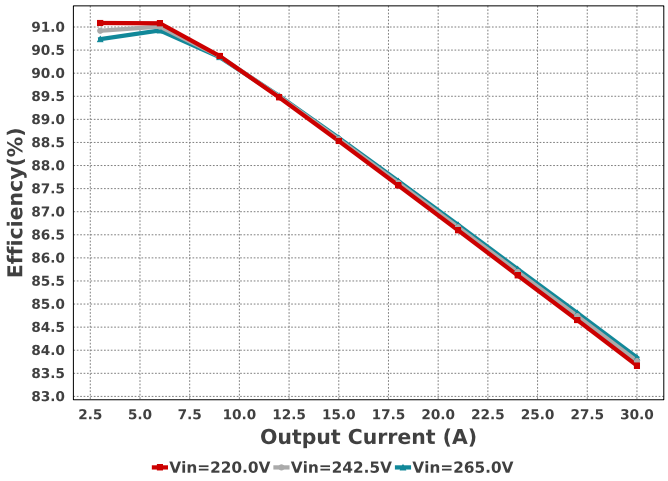
Cin IRMS



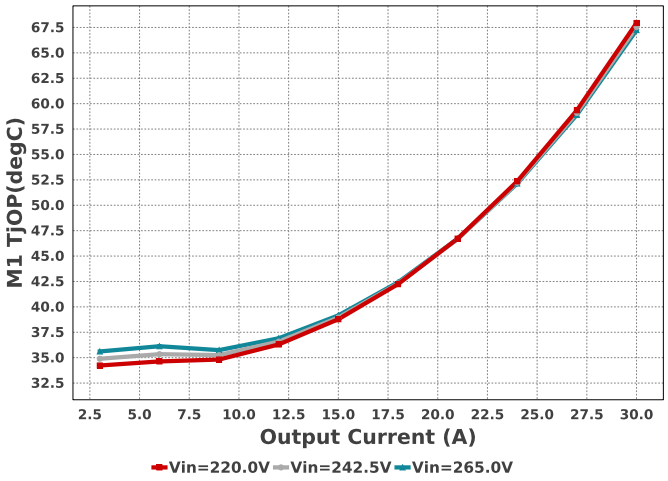
T1 Is1 RMS



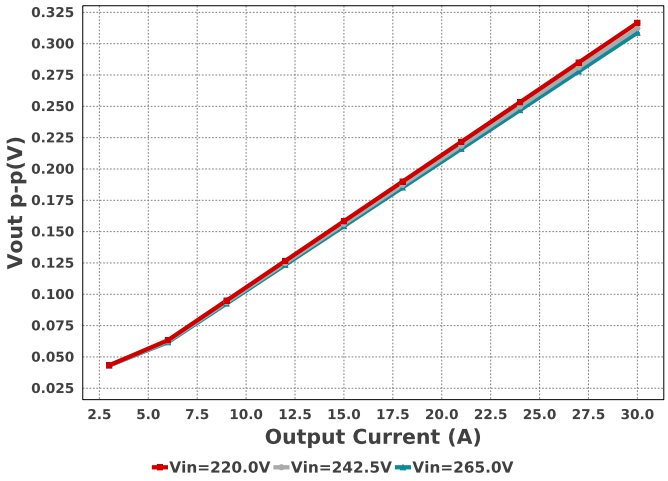
Efficiency



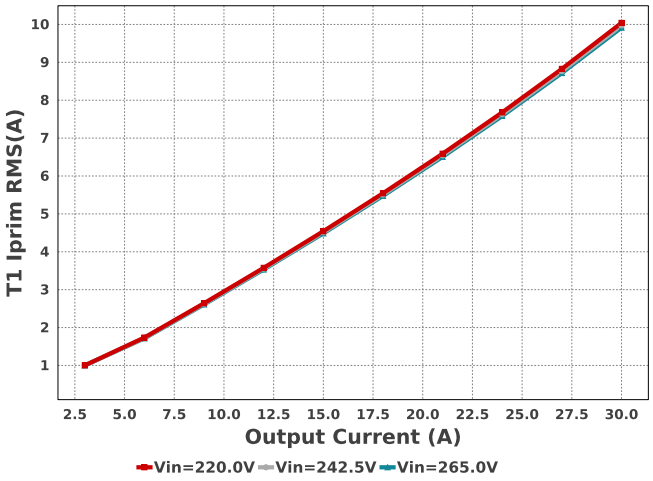
M1 TjOP



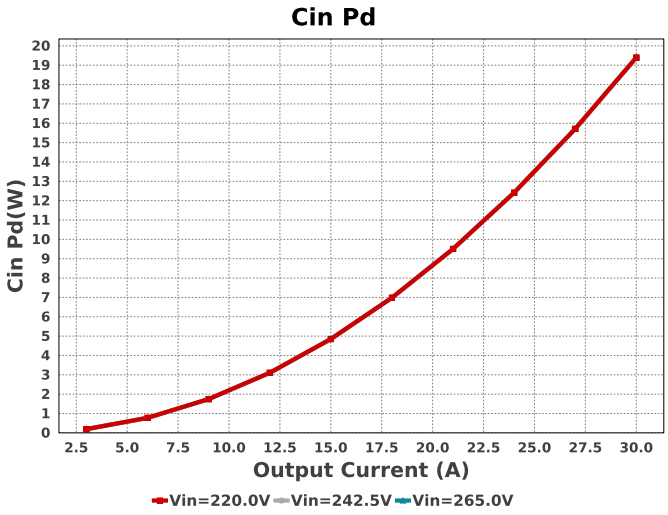
Vout p-p

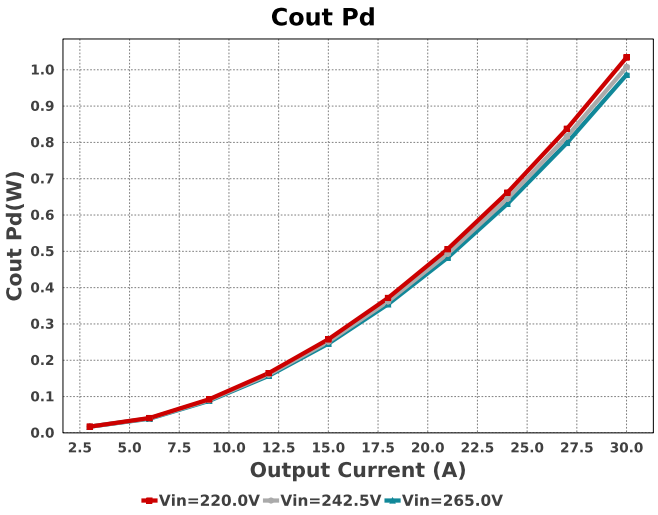


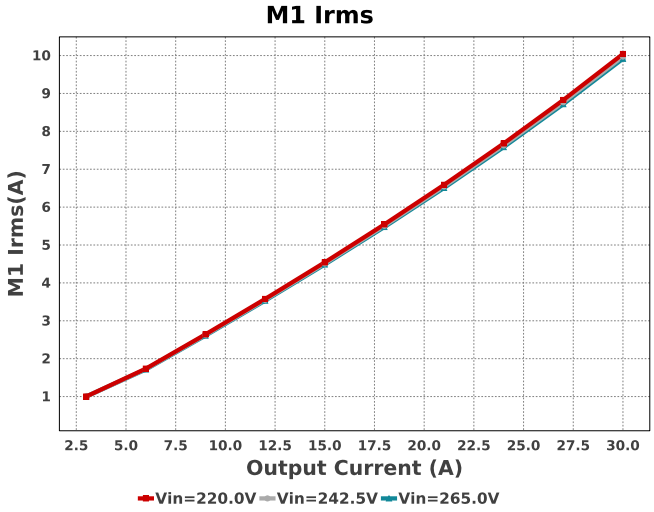
T1 Iprim RMS

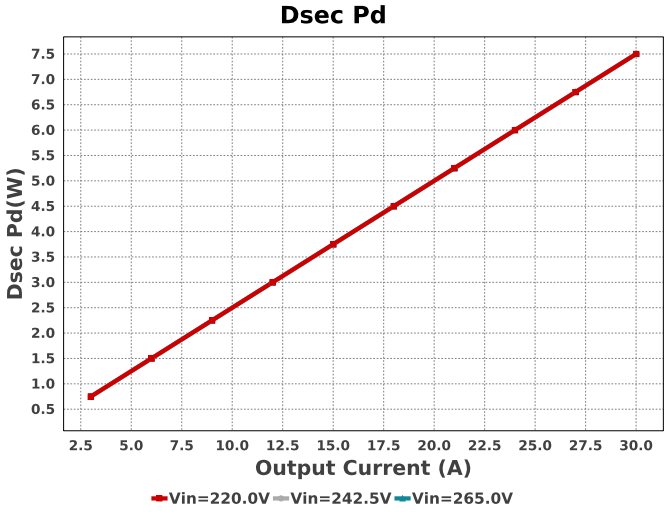


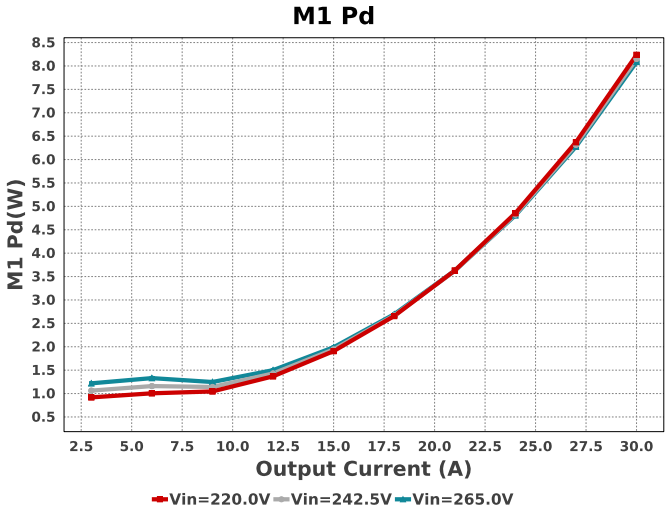
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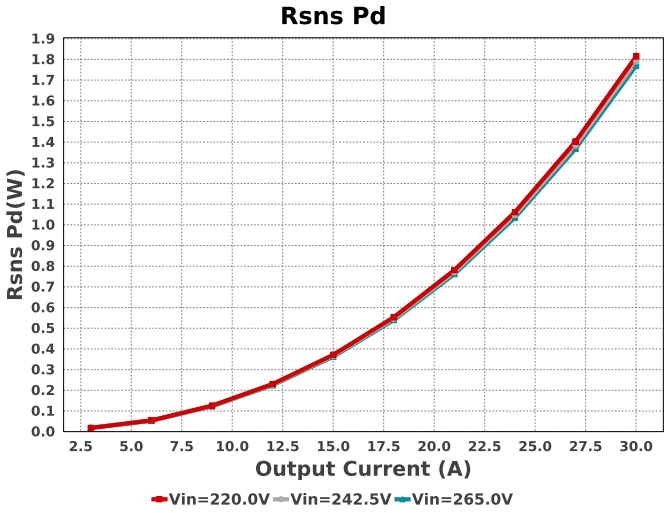




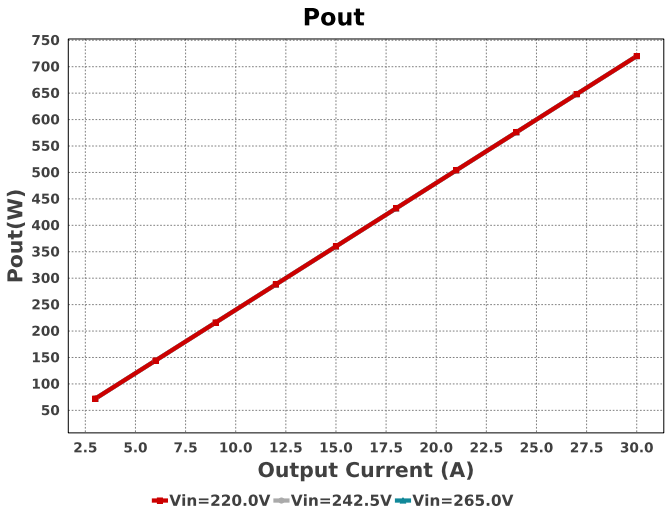


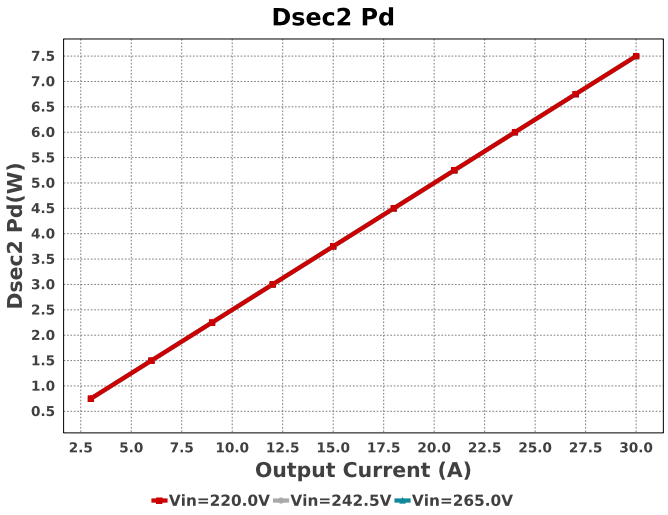


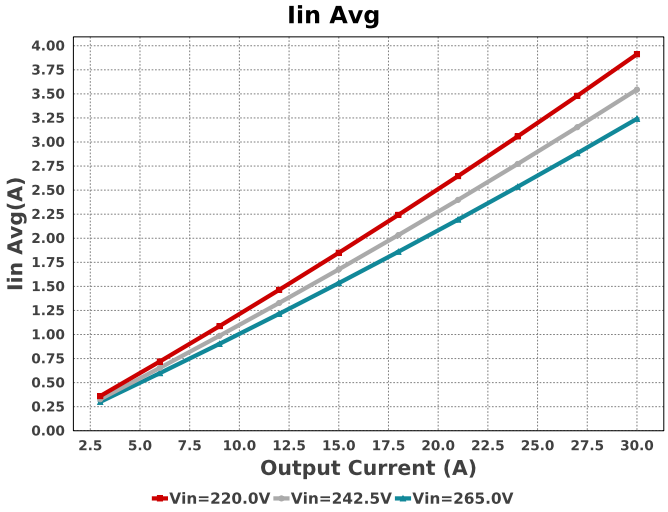


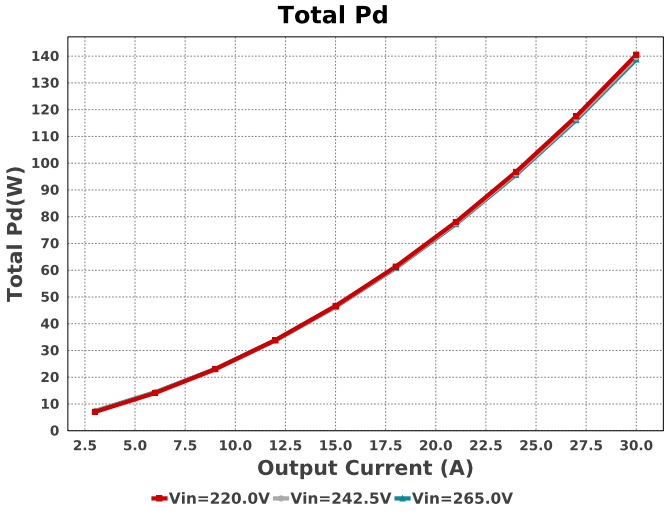


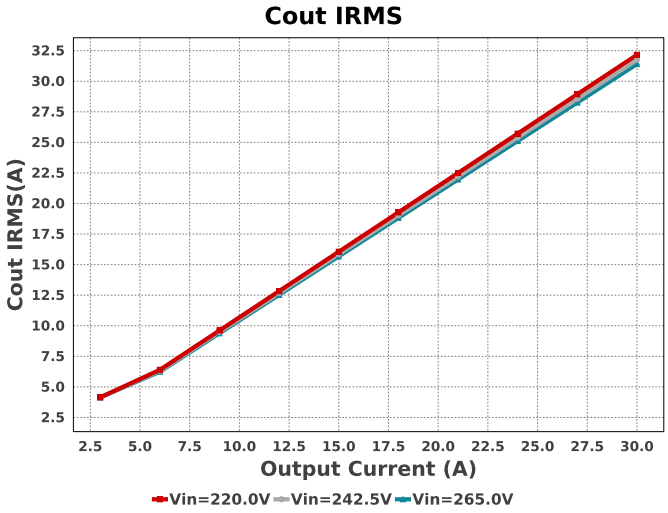
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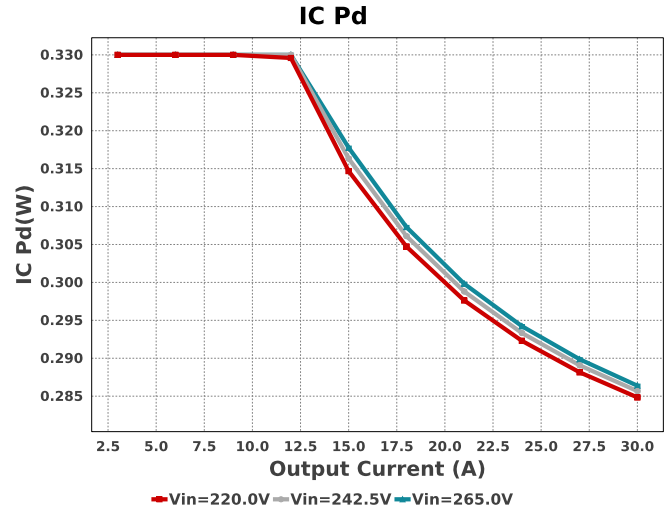




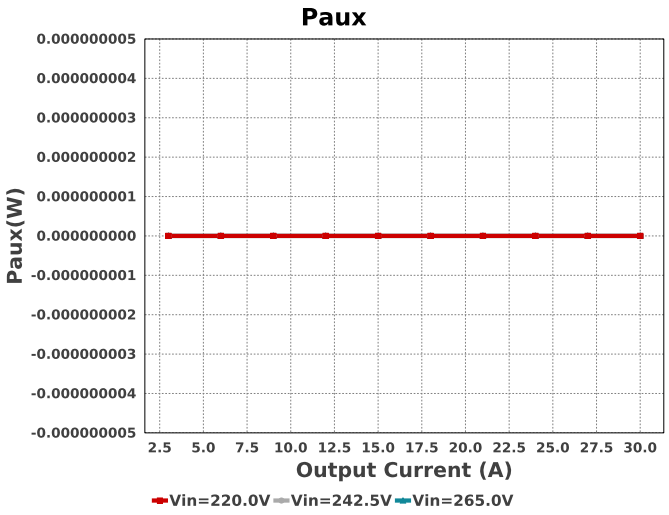


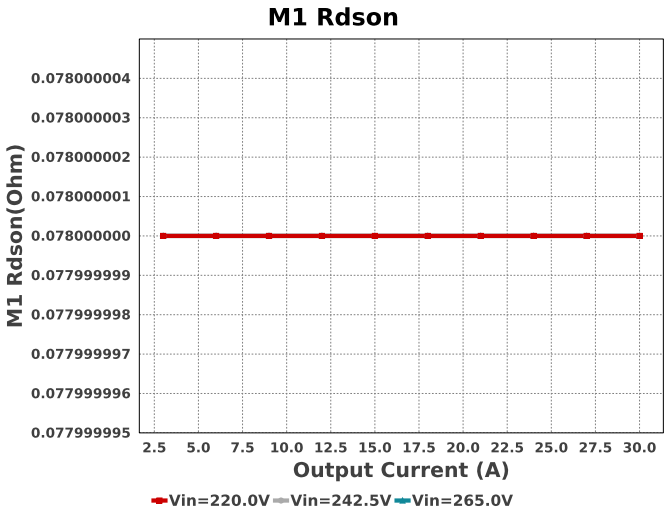


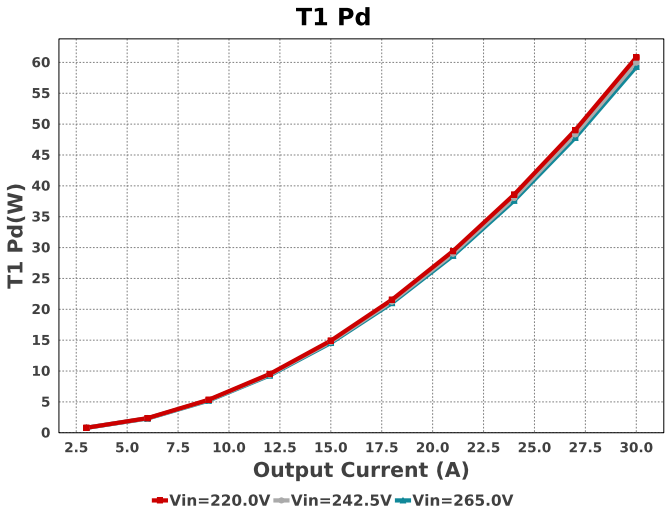




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Operating Values

#	Name	Value	Category	Description
1.	Cin IRMS	8.339 A	Capacitor	Input capacitor RMS ripple current
2.	Cin Pd	19.395 W	Capacitor	Input capacitor power dissipation
3.	Cout IRMS	32.156 A	Capacitor	Output capacitor RMS ripple current
4.	Cout Pd	1.034 W	Capacitor	Output capacitor power dissipation
5.	Dsec Pd	7.5 W	Diode	Secondary Diode Power Dissipation
6.	Dsec2 Pd	7.5 W	Diode	Secondary Diode Power Dissipation
7.	IC Pd	284.84 mW	IC	IC power dissipation
8.	IC Tj	61.019 degC	IC	IC junction temperature
9.	ICThetaJA	108.9 degC/W	IC	IC junction-to-ambient thermal resistance
10.	Iin Avg	3.912 A	IC	Average input current
11.	M1 Irms	10.078 A	Mosfet	Q Iavg
12.	M1 Pd	8.297 W	Mosfet	MOSFET power dissipation
13.	M1 Rdson	78.0 mOhm	Mosfet	Drain-Source On-resistance
14.	M1 TjOP	68.217 degC	Mosfet	M1 MOSFET junction temperature
15.	Cin Pd	19.395 W	Power	Input capacitor power dissipation
16.	Cout Pd	1.034 W	Power	Output capacitor power dissipation
17.	Dsec Pd	7.5 W	Power	Secondary Diode Power Dissipation
18.	Dsec2 Pd	7.5 W	Power	Secondary Diode Power Dissipation
19.	IC Pd	284.84 mW	Power	IC power dissipation
20.	M1 Pd	8.297 W	Power	MOSFET power dissipation
21.	Paux	0.0 W	Power	Power Dissipation in Raux and Daux
22.	Rsns Pd	1.828 W	Power	Current Limit Sense Resistor Power Dissipation
23.	Snubber Pd	27.0 W	Power	Snubber Power Dissipation
24.	T1 Pd	60.858 W	Power	Estimated Losses in Transformer
25.	Total Pd	140.681 W	Power	Total Power Dissipation
26.	Rsns Pd	1.828 W	Resistor	Current Limit Sense Resistor Power Dissipation
27.	BOM Count	51	System Information	Total Design BOM count
28.	Duty Cycle	22.12 %	System Information	Duty cycle
29.	Efficiency	83.656 %	System Information	Steady state efficiency

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#	Name	Value	Category	Description
30.	FootPrint	2.598 k mm^2	System Information	Total Foot Print Area of BOM components
31.	Frequency	24.258 kHz	System Information	Switching frequency
32.	Iout	30.0 A	System Information	Iout operating point
33.	Mode	DCM	System Information	Conduction Mode
34.	Pout	720.06 W	System Information	Total output power
35.	Total BOM	NA	System Information	Total BOM Cost
36.	Vin	220.0 V	System Information	Vin operating point
37.	Vout	24.0 V	System Information	Operational Output Voltage
38.	Vout p-p	316.605 mV	System Information	Peak-to-peak output ripple voltage
39.	Paux	0.0 W	Transformer	Power Dissipation in Raux and Daux
40.	T1 Iprim RMS	10.078 A	Transformer	Transformer Primary RMS Current
41.	T1 Iprim pk	37.113 A	Transformer	Transformer Primary Peak Current
42.	T1 Is1 RMS	44.143 A	Transformer	Transformer Secondary1 RMS Current
43.	T1 Pd	60.858 W	Transformer	Estimated Losses in Transformer

Design Inputs

Name	Value	Description
Iout	30.0	Maximum Output Current
VinMax	265.0	Maximum input voltage
VinMin	220.0	Minimum input voltage
Vout	24.0	Output Voltage
acFrequency	50.0	AC Frequency
base_pn	UCC28600	Base Product Number
source	AC	Input Source Type
Ta	30.0	Ambient temperature

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WEBENCH® Assembly

Component Testing

Some published data on components in datasheets such as Capacitor ESR and Inductor DC resistance is based on conservative values that will guarantee that the components always exceed the specification. For design purposes it is usually better to work with typical values. Since this data is not always available it is a good practice to measure the Capacitance and ESR values of Cin and Cout, and the inductance and DC resistance of L1 before assembly of the board. Any large discrepancies in values should be electrically simulated in WEBENCH to check for instabilities and thermally simulated in WebTHERM to make sure critical temperatures are not exceeded.

Soldering Component to Board

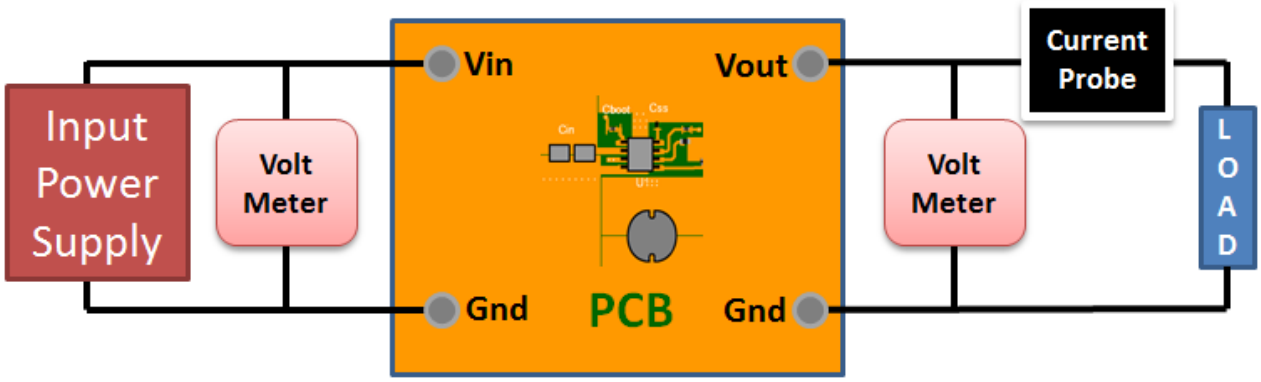
If board assembly is done in house it is best to tack down one terminal of a component on the board then solder the other terminal. For surface mount parts with large tabs, such as the DPAK, the tab on the back of the package should be pre-tinned with solder, then tacked into place by one of the pins. To solder the tab town to the board place the iron down on the board while resting against the tab, heating both surfaces simultaneously. Apply light pressure to the top of the plastic case until the solder flows around the part and the part is flush with the PCB. If the solder is not flowing around the board you may need a higher wattage iron (generally 25W to 30W is enough).

Initial Startup of Circuit

It is best to initially power up the board by setting the input supply voltage to the lowest operating input voltage 220.0V and set the input supply's current limit to zero. With the input supply off connect up the input supply to Vin and GND. Connect a digital volt meter and a load if needed to set the minimum Iout of the design from Vout and GND. Turn on the input supply and slowly turn up the current limit on the input supply. If the voltage starts to rise on the input supply continue increasing the input supply current limit while watching the output voltage. If the current increases on the input supply, but the voltage remains near zero, then there may be a short or a component misplaced on the board. Power down the board and visually inspect for solder bridges and recheck the diode and capacitor polarities. Once the power supply circuit is operational then more extensive testing may include full load testing, transient load and line tests to compare with simulation results.

Load Testing

The setup is the same as the initial startup, except that an additional digital voltmeter is connected between Vin and GND, a load is connected between Vout and GND and a current meter is connected in series between Vout and the load. The load must be able to handle at least rated output power + 50% (7.5 watts for this design). Ideally the load is supplied in the form of a variable load test unit. It can also be done in the form of suitably large power resistors. When using an oscilloscope to measure waveforms on the prototype board, the ground leads of the oscilloscope probes should be as short as possible and the area of the loop formed by the ground lead should be kept to a minimum. This will help reduce ground lead inductance and eliminate EMI noise that is not actually present in the circuit.



Design Assistance

1. Quasi-Resonant Flyback Green-Mode Controller
2. Master key : CD3AC80E598892CF1F32BD0F49DB9A50[v1]
3. **UCC28600** Product Folder : http://www.ti.com/product/UCC28600 : contains the data sheet and other resources.

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